



TASK 1 – DBMS PROJECT

GROUP 7

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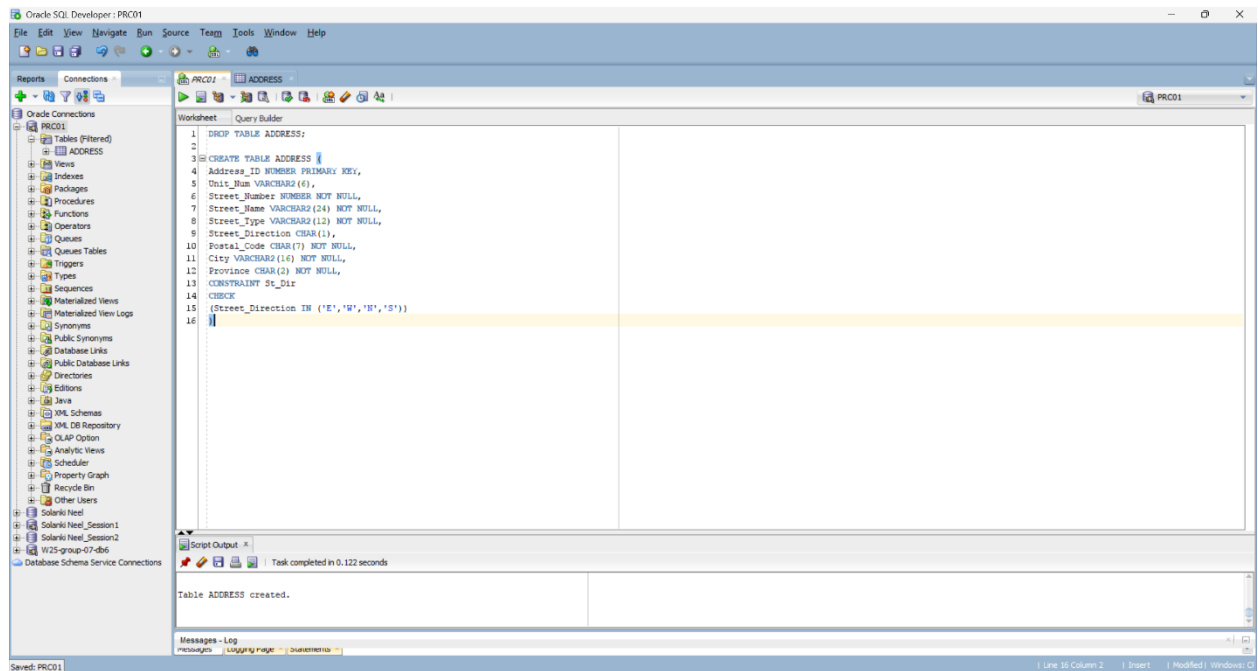
TASK 1

TASK DISTRIBUTION

NEEL PARESH SOLANKI	Designed and implemented the Talend job designs, including metadata connections and data transformation logic.
VIMAL SHANTIBHAI LAKHANI	Set up the Oracle database using SQL Developer and created the address table.
PRATHAM MILANKUMAR DOSHI	Prepared this Project Document and helped in step-by-step implementation
RAKSHIT KHANNA	Provided support during the project by assisting with test runs and helping ensure the overall process went smoothly

Creating Tables and Connections in Oracle SQL Developer for Task 1

1. Creating Address Table



```
CREATE TABLE ADDRESS (  
    Address_ID NUMBER PRIMARY KEY,  
    Unit_Num VARCHAR2(6),  
    Street_Number NUMBER NOT NULL,  
    Street_Name VARCHAR2(24) NOT NULL,  
    Street_Type VARCHAR2(12) NOT NULL,  
    Street_Direction CHAR(1),  
    Postal_Code CHAR(7) NOT NULL,  
    City VARCHAR2(16) NOT NULL,  
    Province CHAR(2) NOT NULL,  
    CONSTRAINT St_Dir  
    CHECK  
    (Street_Direction IN ('E','W','N','S'))  
)
```

2. Created ORACLE CONNECTION Named PRC01

The screenshot shows the 'New / Select Database Connection' dialog box. On the left, a table lists existing connections. The main area is configured for a new connection named 'PRC01'.

Connection Name	Connection Details
PRC01	prc@//db6.fast.s...
Solanki Neel	s4_Solannee@//...
Solanki Neel_Ses...	s4_Solannee@//...
Solanki Neel_Ses...	s4_Solannee@//...
W25-group-07-db6	SYSTEM@//db6.f...

Name: PRC01

Database Type: Oracle

User Info: Proxy User

Authentication Type: Default

Username: prc

Role: default

Password: [masked]

Save Password: ☒

Connection Type: Basic

Details: Advanced

Hostname: db6.fast.sheridanc.on.ca

Port: 1521

Service name: U07.World

Status:

Buttons: Help, Save, Clear, Test, Connect, Cancel

3. Created ORACLE CONNECTION Named W25-group-07-db6

The screenshot shows the 'New / Select Database Connection' dialog box. On the left, a table lists existing connections. The main area is configured for a new connection named 'W25-group-07-db6'.

Connection Name	Connection Details
PRC01	prc@//db6.fast.s...
Solanki Neel	s4_Solannee@//...
Solanki Neel_Ses...	s4_Solannee@//...
Solanki Neel_Ses...	s4_Solannee@//...
W25-group-07-db6	SYSTEM@//db6.f...

Name: W25-group-07-db6

Database Type: Oracle

User Info: Proxy User

Authentication Type: Default

Username: SYSTEM

Role: default

Password: [masked]

Save Password: ☒

Connection Type: Basic

Details: Advanced

Hostname: db6.fast.sheridanc.on.ca

Port: 1521

Service name: U07.World

Status:

Buttons: Help, Save, Clear, Test, Connect, Cancel

Creating Metadata DB Connection in Talend to connect with Oracle Database

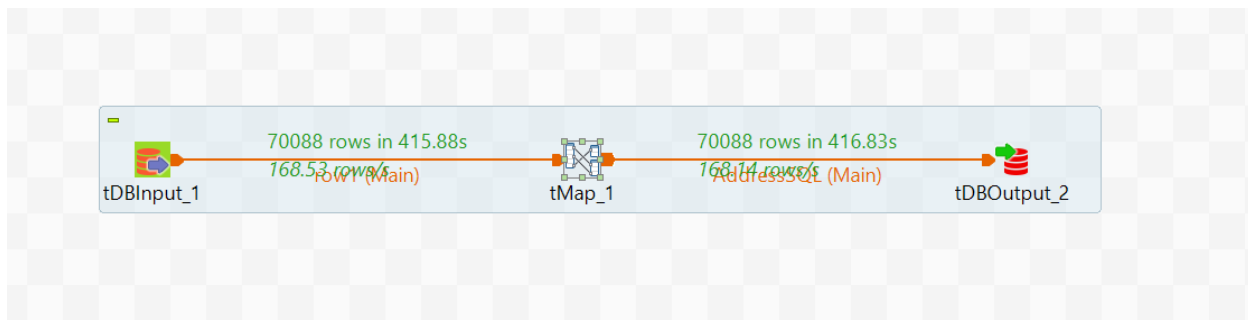
1. Created DB connection in Metadata using **MICROSOFT SQL SERVER** and named as **U07.World**

The screenshot shows the 'Database Connection' dialog box in Talend, titled 'Update Database Connection - Step 2/2'. It includes a warning icon and text: 'You must press the Check Button to check the Database Setting'. The 'DB Type' is set to 'Microsoft SQL Server'. The 'Db Version' is 'Open source JTDS'. The 'String of Connection' is 'jdbc:tds:sqlserver://dbr.fast.sheridanc.on.ca:1433/Integration;'. The 'Login' is 'DataIntegrator', 'Password' is masked with dots, 'Server' is 'dbr.fast.sheridanc.on.ca', 'Port' is '1433', 'DataBase' is 'Integration', and 'Schema' is 'dbo'. There is an empty 'Additional parameters' field. A 'Test connection' button is at the bottom right. At the bottom are '< Back', 'Next >', 'Finish', and 'Cancel' buttons. A link 'How to install a driver' is at the bottom left.

2. Created DB connection in Metadata using **ORACLE WITH SERVICE NAME** and named as **AddressOracle**.

The screenshot shows the 'Database Connection' dialog box in Talend, titled 'Update Database Connection - Step 2/2'. It includes a warning icon and text: 'You must press the Check Button to check the Database Setting'. The 'DB Type' is set to 'Oracle with service name'. The 'Db Version' is 'Oracle 12'. The 'String of Connection' is 'jdbc:oracle:thin:@(description=(address=(protocol=tcp)(host=db6.fast.sheridanc.on.ca)(port=1521))(connect_data={service_name=U07.World})).'. The 'Login' is 'PRC', 'Password' is masked with dots, 'Server' is 'db6.fast.sheridanc.on.ca', 'Port' is '1521', 'Service name' is 'U07.World', and 'Schema' is 'PRC'. There is an empty 'Additional parameters' field. A 'Test connection' button is at the bottom right. Below it are 'Export as context' and 'Revert Context' buttons. At the bottom are '< Back', 'Next >', 'Finish', and 'Cancel' buttons. A link 'How to install a driver' is at the bottom left.

3. Job Design Connection



Here In this Connection:

- tDBInput_1**: Reads data from a database table (Address Table) and sends 70,088 rows to the next component.
- tMap_1**: Transforms and maps the incoming data from tDBInput_1 to the output structure.
- tDBOutput_2**: Writes the transformed data into a destination database table.

4. Detailed tMap

The screenshot displays the Talend Open Studio for Data Integration interface, specifically the configuration for the **tMap_1** component. The interface is divided into several panes:

- Schema editor**: Shows the input schema for **row1** with columns: STREET_NUM, UNIT, STREET_NAME, STREET_TYPE, STREET_DIR, POSTAL_CODE, CITY, and PROVINCE.
- Expression editor**: Displays the mapping logic for the output schema. It includes a **Var** section with expressions like `new BigDecimal(row1.STREET_NUM)` and `row1.STREET_NAME`, and a **Column** section with expressions like `row1.STREET_NUM` and `row1.STREET_NAME`.
- Auto map**: Shows the output schema for **AddressSQL** with columns: ADDRESS_ID, UNIT_NUM, STREET_NUMBER, STREET_NAME, STREET_DIRECTION, POSTAL_CODE, CITY, and PROVINCE.
- Schema editor**: Shows the output schema for **AddressSQL** with columns: ADDRESS_ID, UNIT_NUM, STREET_NUMBER, STREET_NAME, STREET_DIRECTION, POSTAL_CODE, CITY, and PROVINCE.

The **Schema editor** for **row1** and **AddressSQL** is shown below:

Column	K...	Type	N	Date Pattern (Ctrl+Spa...	Length	Precision	Default	Comment
STREET_NUM		String			6	0		
UNIT		String			10	0		
STREET_NAME		String			24	0		
STREET_TYPE		String			12	0		
STREET_DIR		String			4	0		
POSTAL_CODE		String			6	0		
CITY		String			24	0		
PROVINCE		String			2	0		

Column	K...	Type	N	Date Pattern (Ctrl+Spa...	Length	Precision	Default	Comment
ADDRESS_ID		Integer			0			
UNIT_NUM		String			10	0		
STREET_NUMBER		String			6			
STREET_NAME		String			24	0		
STREET_TYPE		String			12	0		
STREET_DIRECTION		String			4	0		
POSTAL_CODE		String			6	0		
CITY		String			24	0		
PROVINCE		String			2	0		

5. Job Design Success for fetching Address Rows

The screenshot shows the Talend Open Studio interface. The left pane displays the 'Repository' tree with the job 'Job AddressSQL 0.1' selected. The main workspace shows the job design with three components: tDBInput_1, tMap_1, and tDBOutput_2. The tMap_1 component is configured to map data from the tDBInput_1 connector to the tDBOutput_2 connector. The job is named 'Job AddressSQL 0.1'. The right pane shows the 'Palette' with various connectors and components.

Job AddressSQL 0.1

Execution

```
Starting job AddressSQL at 21:29 21-03-2025.  
[statistic] connecting to socket on port 4025  
[statistic] connected  
[statistic] disconnected  
Job AddressSQL ended at 21:36 21-03-2025. [Exit code = 0]
```

6. Final Address Table with all Values Inserted

The screenshot shows the Oracle SQL Developer interface. The left pane displays the 'Reports' tree with the table 'ADDRESS' selected. The main workspace shows the 'Query Result' window with the data from the 'ADDRESS' table. The table has 29 rows of data. The columns are ADDRESS_ID, UNIT_NUM, STREET_NAME, STREET_TYPE, STREET_DIRECTION, POSTAL_CODE, CITY, and PROVINCE.

ADDRESS_ID	UNIT_NUM	STREET_NAME	STREET_TYPE	STREET_DIRECTION	POSTAL_CODE	CITY	PROVINCE
1	454 (null)	2349 CHEVERIE	ST	(null)	000000	GARVILLE OH	
2	455 (null)	2354 CHEVERIE	ST	(null)	000000	GARVILLE OH	
3	456 (null)	2359 CHEVERIE	ST	(null)	000000	GARVILLE OH	
4	457 (null)	2364 CHEVERIE	ST	(null)	000000	GARVILLE OH	
5	458 (null)	2366 CHEVERIE	ST	(null)	000000	GARVILLE OH	
6	459 (null)	2369 CHEVERIE	ST	(null)	000000	GARVILLE OH	
7	460 (null)	2369 CHEVERIE	ST	(null)	000000	GARVILLE OH	
8	461 (null)	2371 CHEVERIE	ST	(null)	000000	GARVILLE OH	
9	462 (null)	2373 CHEVERIE	ST	(null)	000000	GARVILLE OH	
10	463 (null)	2375 CHEVERIE	ST	(null)	000000	GARVILLE OH	
11	464 (null)	2384 BACCARO	RD	(null)	000000	GARVILLE OH	
12	465 (null)	2364 DEVON	RD	(null)	000000	GARVILLE OH	
13	466 (null)	2366 DEVON	RD	(null)	000000	GARVILLE OH	
14	467 (null)	2368 DEVON	RD	(null)	000000	GARVILLE OH	
15	468 (null)	2370 DEVON	RD	(null)	000000	GARVILLE OH	
16	469 (null)	2372 DEVON	RD	(null)	000000	GARVILLE OH	
17	470 (null)	2374 DEVON	RD	(null)	000000	GARVILLE OH	
18	471 (null)	2376 CHEVERIE	ST	(null)	000000	GARVILLE OH	
19	472 (null)	2370 CHEVERIE	ST	(null)	000000	GARVILLE OH	
20	473 (null)	2369 BACCARO	RD	(null)	000000	GARVILLE OH	
21	474 (null)	2364 BACCARO	RD	(null)	000000	GARVILLE OH	
22	475 (null)	2360 BACCARO	RD	(null)	000000	GARVILLE OH	
23	476 (null)	2356 BACCARO	RD	(null)	000000	GARVILLE OH	
24	477 (null)	2352 BACCARO	RD	(null)	000000	GARVILLE OH	
25	478 (null)	2363 CHEVERIE	ST	(null)	000000	GARVILLE OH	
26	479 (null)	2357 CHEVERIE	ST	(null)	000000	GARVILLE OH	
27	480 (null)	2353 CHEVERIE	ST	(null)	000000	GARVILLE OH	
28	481 (null)	2347 CHEVERIE	ST	(null)	000000	GARVILLE OH	
29	482 (null)	2343 CHEVERIE	ST	(null)	000000	GARVILLE OH	

7. Changes Made in TMap to solve the error I was getting before, Inserted some Query.

Integration - tMap - tMap_1

Find :   

Var	Expression	Type	Variable
	new BigDecimal(row1.STREET_NUM)	BigDecimal	☐ var1
	row1.UNIT	String	☐ var2
	row1.STREET_NAME	String	☐ var3
	row1.STREET_TYPE == null row1.STREET_TYPE.isEmpty()? "UNKNOWN" : row1.STREET_TYPE	String	☐ var4
	row1.STREET_DIR	String	☐ var5
	row1.POSTAL_CODE == null row1.POSTAL_CODE.isEmpty()? "000000" : row1.POSTAL_CODE	String	☐ var6
	row1.CITY	String	☐ var7
	row1.PROVINCE	String	☐ var8
	Numeric.sequence("SEQ_ADDRESS_ID",1,1)	int	☐ var9

1. **POSTAL_CODE** = row1.POSTAL_CODE == **null** || row1.POSTAL_CODE.isEmpty()?
"000000" : row1.POSTAL_CODE
2. **ADDRESS_ID** = Numeric.sequence("SEQ_ADDRESS_ID",1,1)
3. **STREET_TYPE** = row1.STREET_TYPE == null || row1.STREET_TYPE.isEmpty()?
"UNKNOWN" : row1.STREET_TYPE